

ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 13: Digital to Time Converter (DTC)

Yiwu Tang



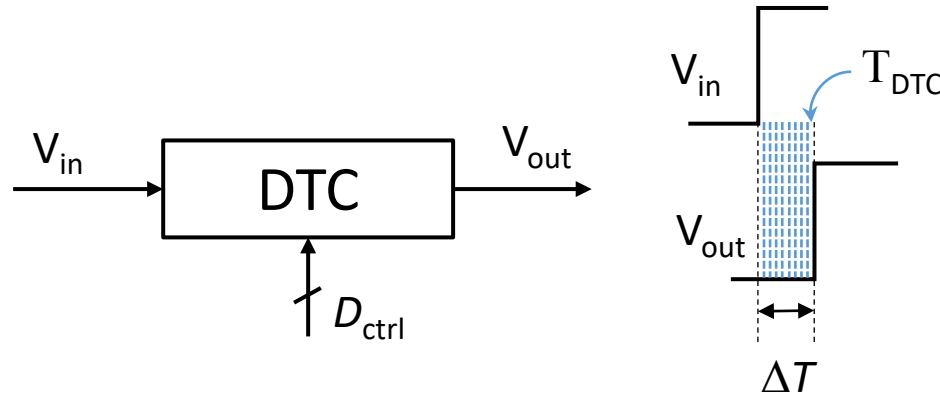
Outline

- What is digital to time converter (DTC)
- Applications and benefits of DTC
- DTC implementations
- DTC performance metrics
 - Resolution
 - Range
 - Noise performance
 - Linearity
- DTC gain and linearity calibration



Digital to Time Converter (DTC)

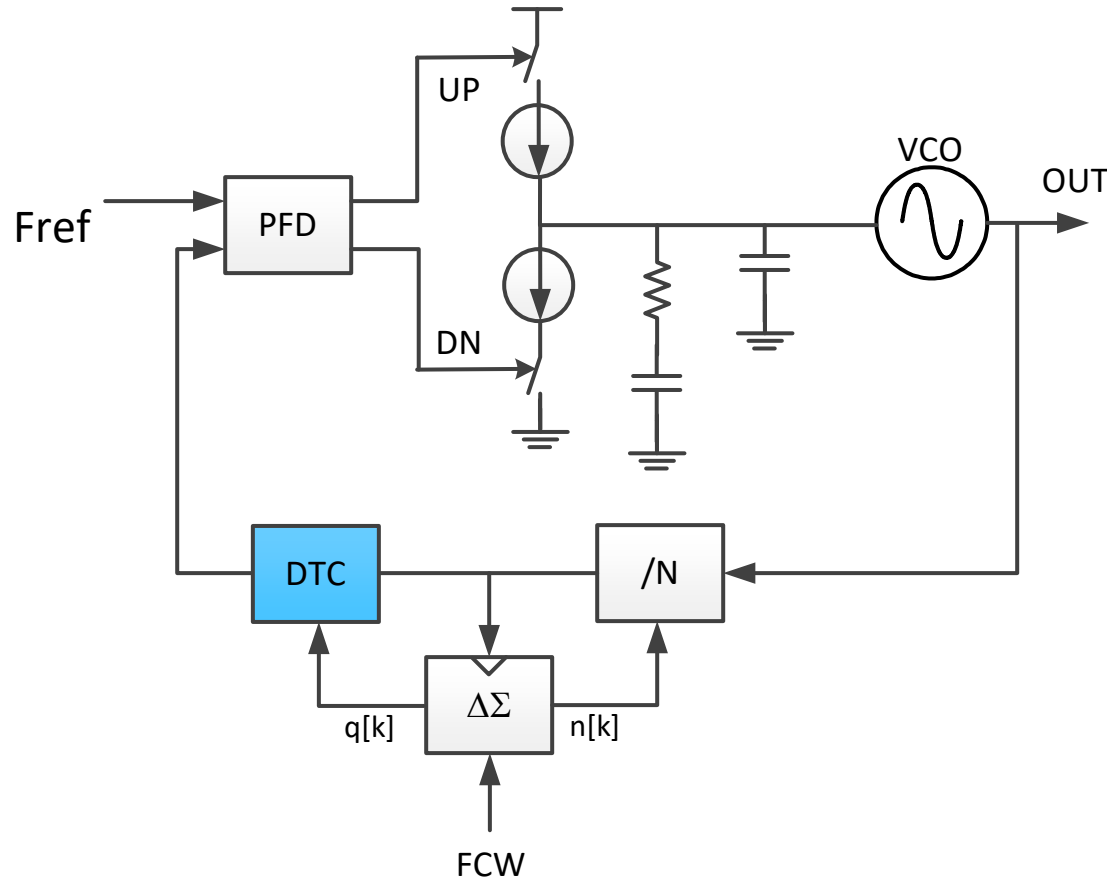
- A Digital-to-time converter (DTC) generates a time delayed version of its input signal based on a digital control code.
- Delta time between the input and output signals $\Delta T = T_0 + T_{\text{res}} \cdot D_{\text{ctrl}}$



- DTC applications
 - Reduce DSM quantization noise in fractional-N PLL, both digital or analog PLL
 - Provide fractional-N functionality in sampling/subsampling PLL and BBPD PLL



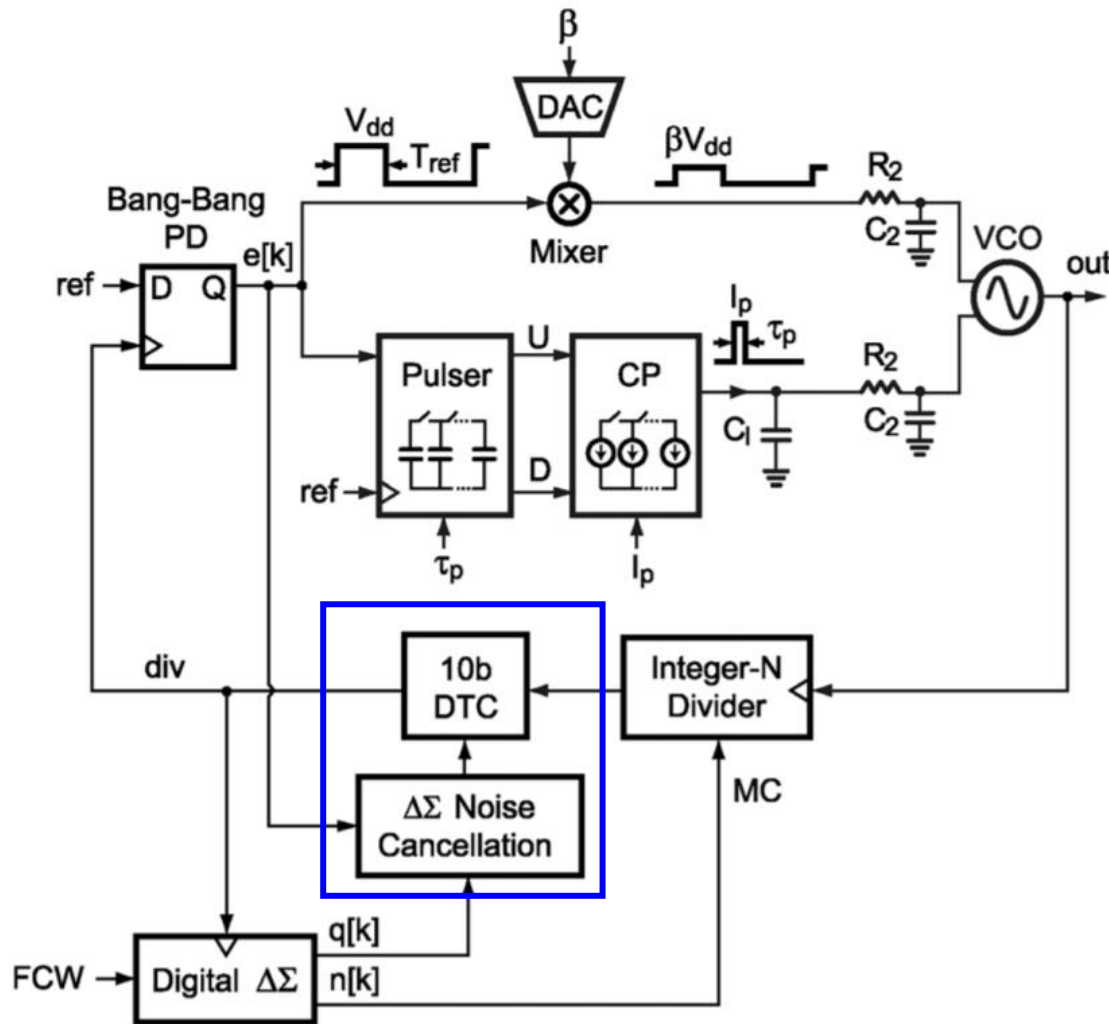
DTC in Analog Fractional-N PLL



- F_v is delayed by high resolution DTC to match F_r phase
- $F_{ref} - F_v$ phase error reduced from $n \cdot T_{VCO}$ (sub-ns) to DTC resolution (ps)
- DSM noise is reduced, loop bandwidth can be widened
- DTC linearity is important to minimize DSM noise folding
- DTC gain accuracy is important and needs to be calibrated



DTC in BBPLL for Fractional-N Frequency Synthesis

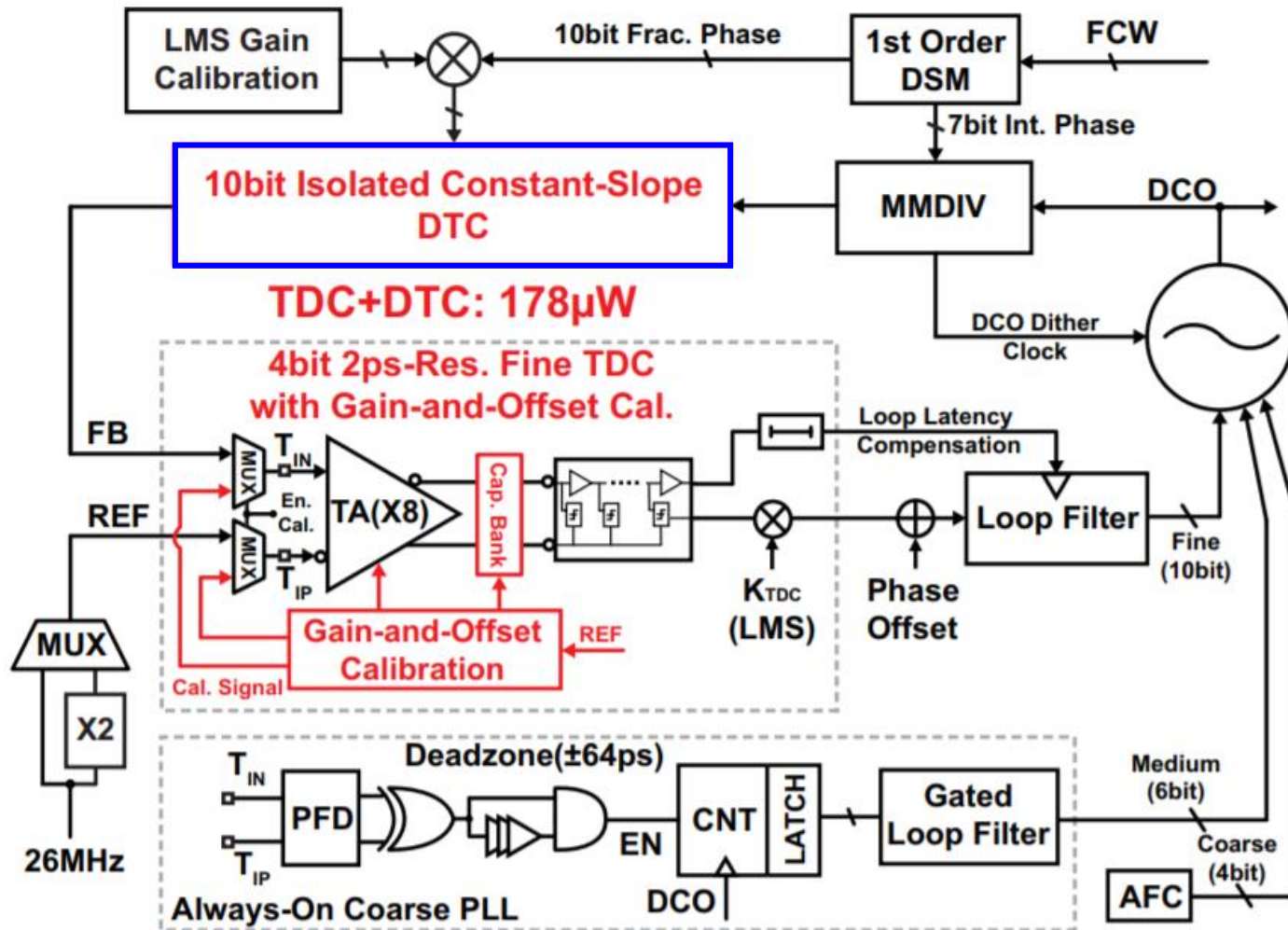


- BBPLL uses BBPD (bang-bang phase detector, aka 1-bit TDC)
 - BBPD output is +1 if Ref leading, -1 if Div leading
- BBPLL can be both APLL or DPLL
- BBPLL is Integer-N PLL
 - BBPD detects only phase difference polarity, not magnitude
- In BBPLL steady locking state, BBPD output average is zero
- To implement fractional-N synthesis, DTC is used to align F_{ref} and F_{div} phases based on DSM quantization noise
 - DTC can be used on either Ref or Div signal

Ref [7]



DTC in Fractional-N ADPLL

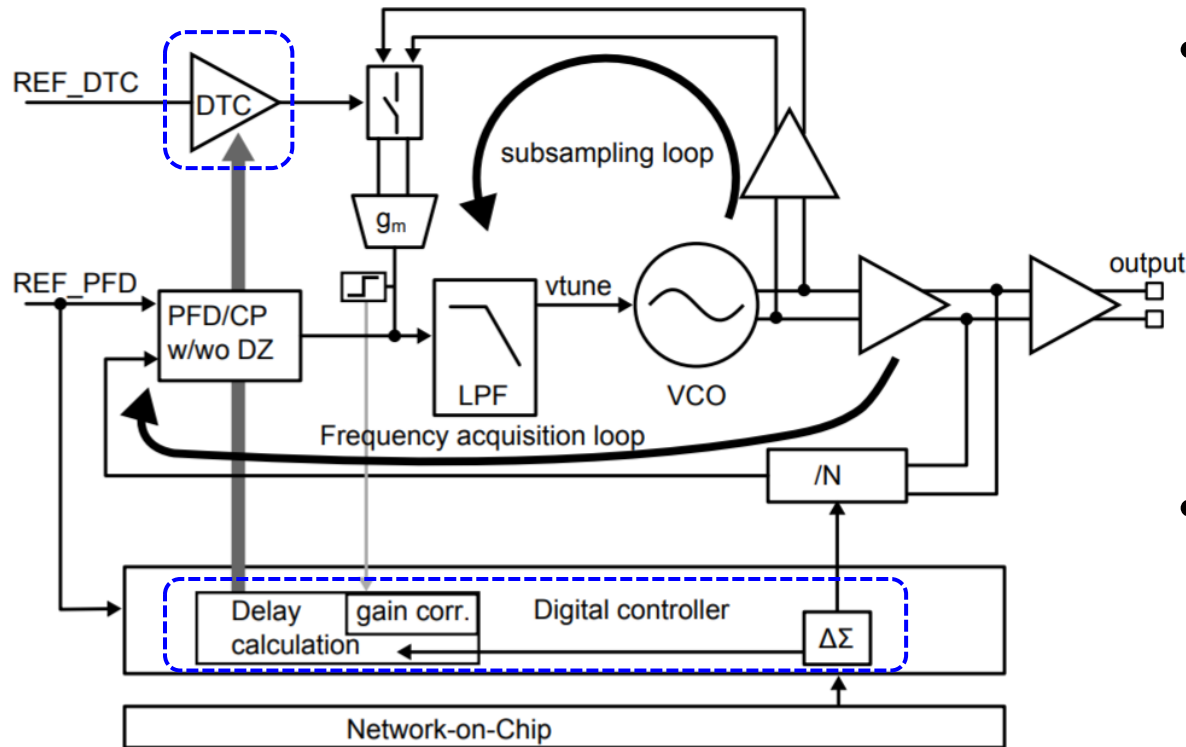


- ADPLL with TDC in feedforward path
- TDC needs wide range to cover DSM range and fine resolution for phase noise performance
 - A few T_{DCO} $10^2\text{ps}\sim\text{ns}$ range
 - Sub-ps resolution
- Use DTC to bring F_v signal phase closer to F_r phase
 - Reduce TDC range requirement
 - Focus on resolution
 - Save power

Ref [6]



DTC in Sub-Sampling PLL

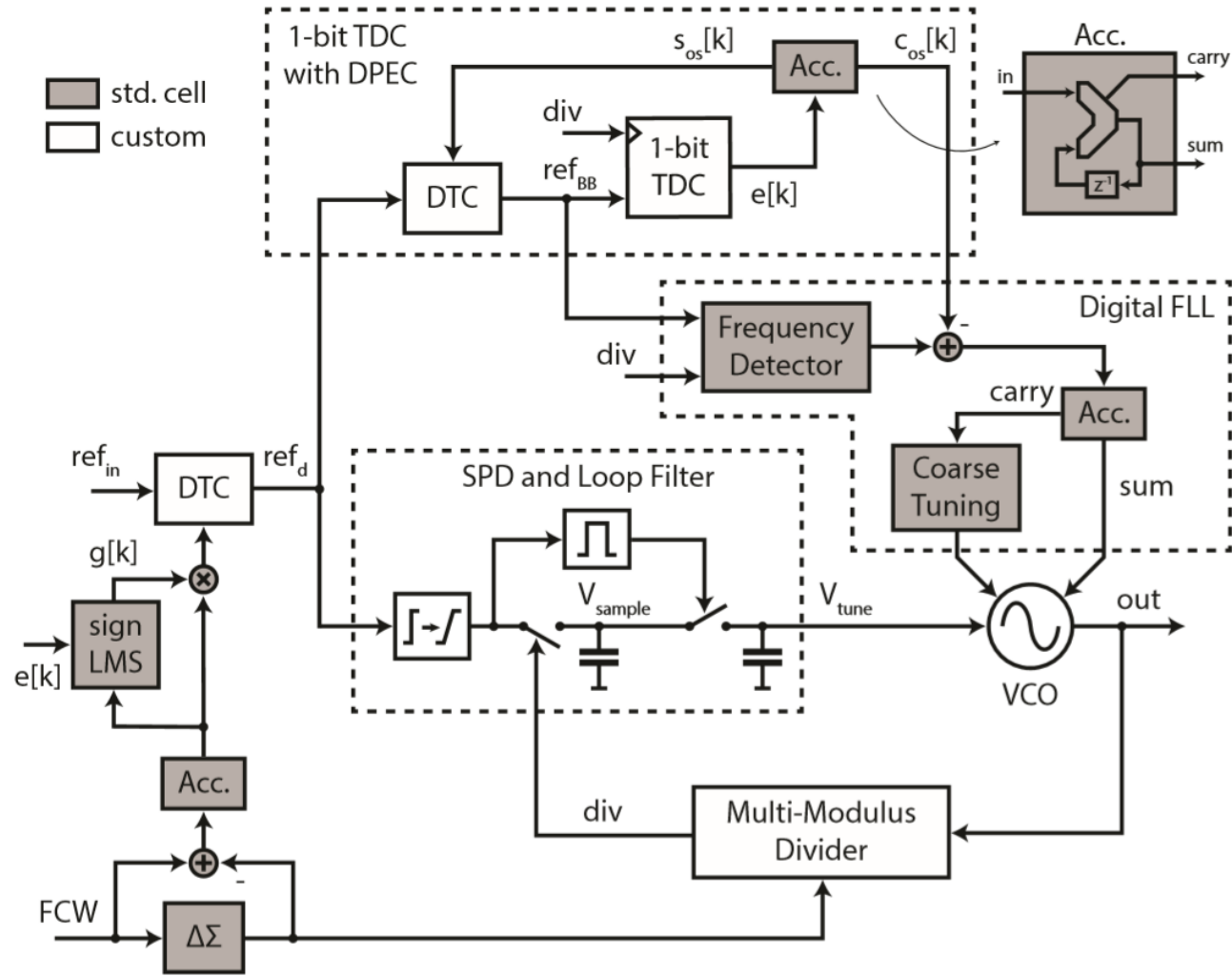


- Sub-sampling PLL is an integer PLL
 - Sampling at the VCO zero crossing with reference signal
 - Reference period is integer multiples of VCO period to maintain zero crossing sampling
 - Integer PLL
- In fractional-N PLL, VCO zero crossing is not always aligned with F_r .
 - F_v is not aligned with F_r at every edge
 - DTC moves F_r to align with F_v edge that aligns F_r to VCO zero crossing
 - DTC delay needed for alignment is determined by instantaneous N divide ratio (DSM output)

Ref [13]



DTC in Sampling PLL

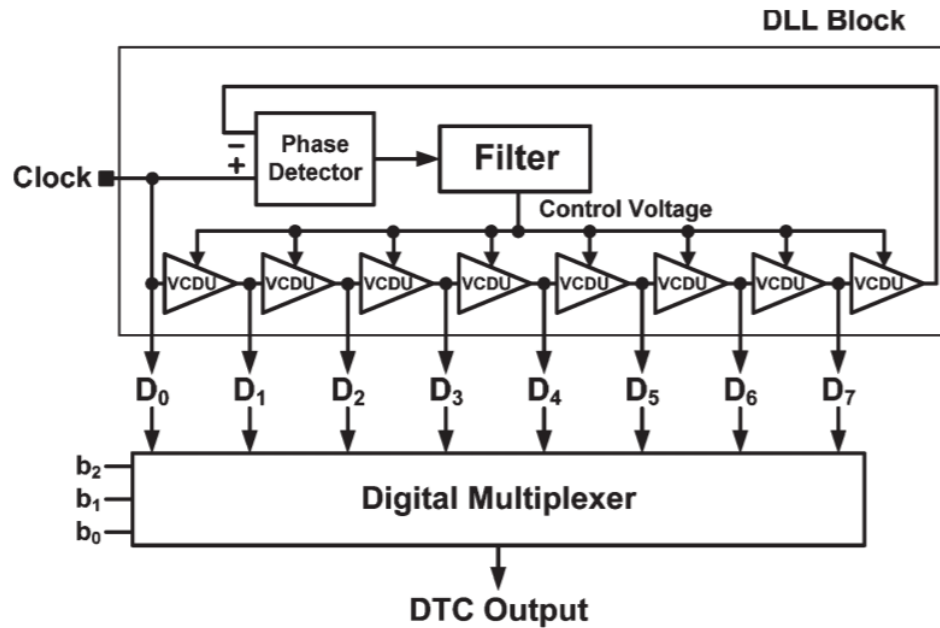


- Sampling PLL with divider in FB path
- Divider controlled by DSM for fractional frequency synthesis
- DTC to align the reference edge with the divided signal edge
- Analog proportional path: type-I PLL
- 1-bit TDC digital integration path: control input phase error range and generate DTC code

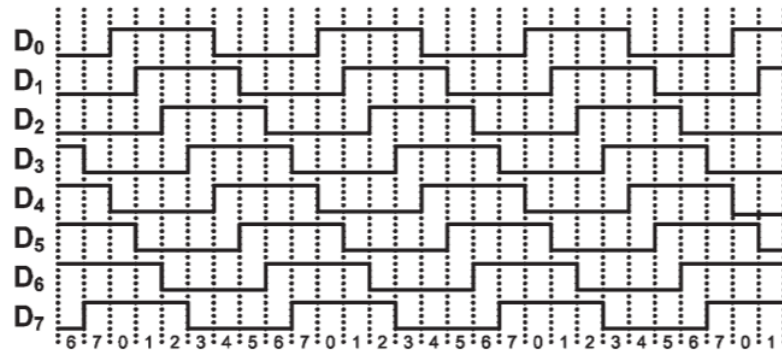
Ref: ISSCC 2020 paper 17.5



DTC Implementation – Delay Line Based



(a)



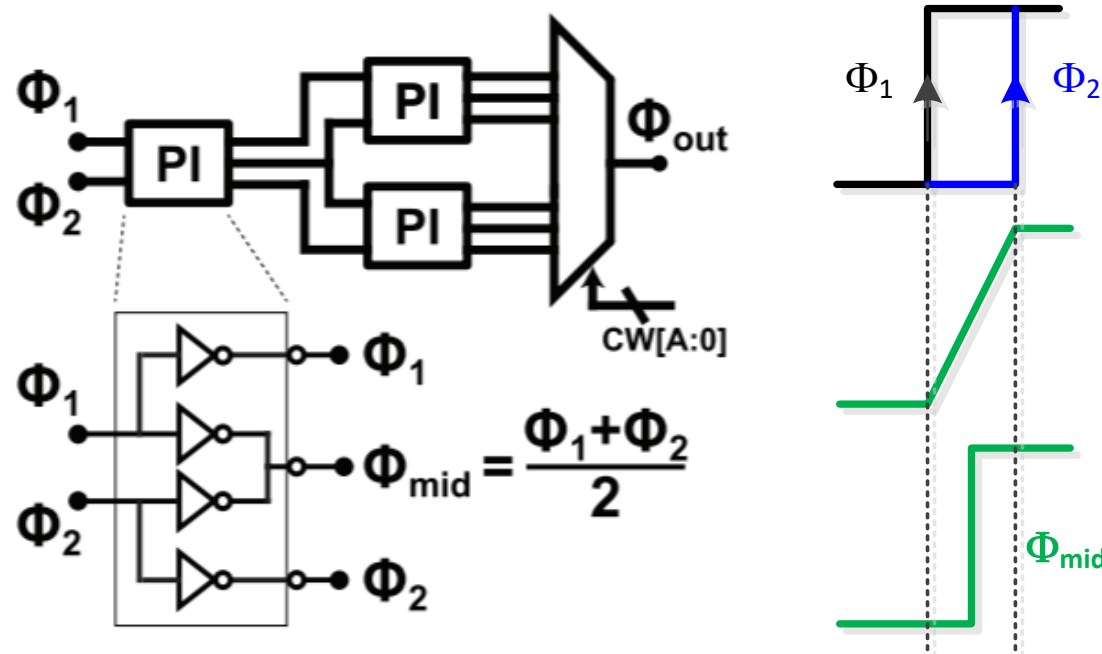
(b)

- Delay line consists of a path of voltage controlled delay units.
- Delay locked loop ensure the total delay of the delay line equals to one input clock -> DTC gain calibration
- DTC resolution determined by unit delay time, technology dependent
- DTC linearity determined by unit delay mismatch

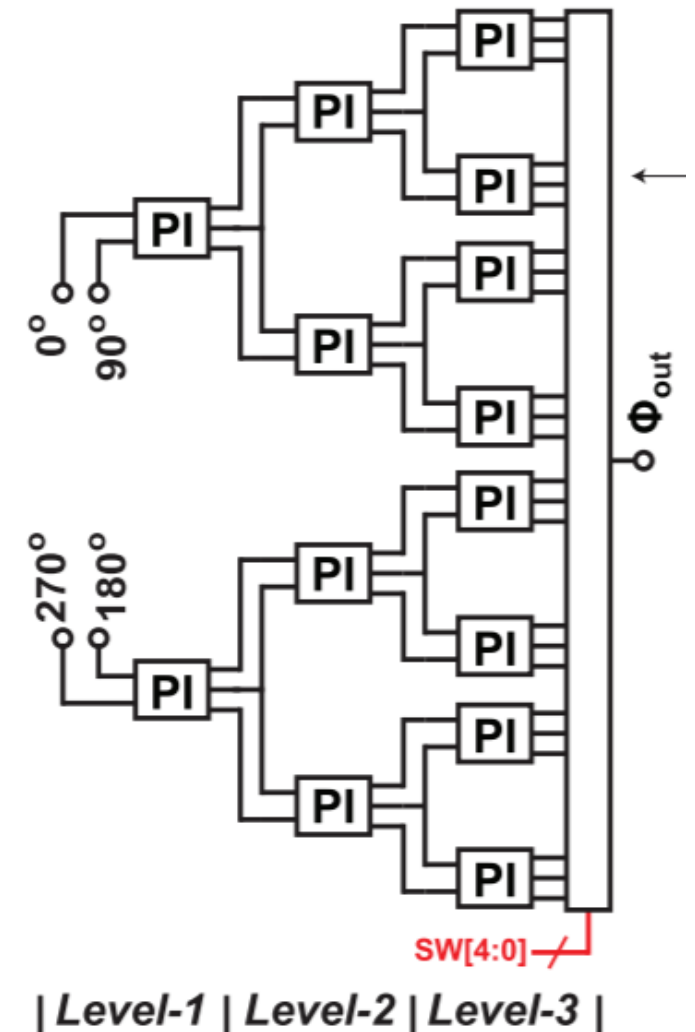
Ref [8]



DTC Implementation – Phase Interpolation

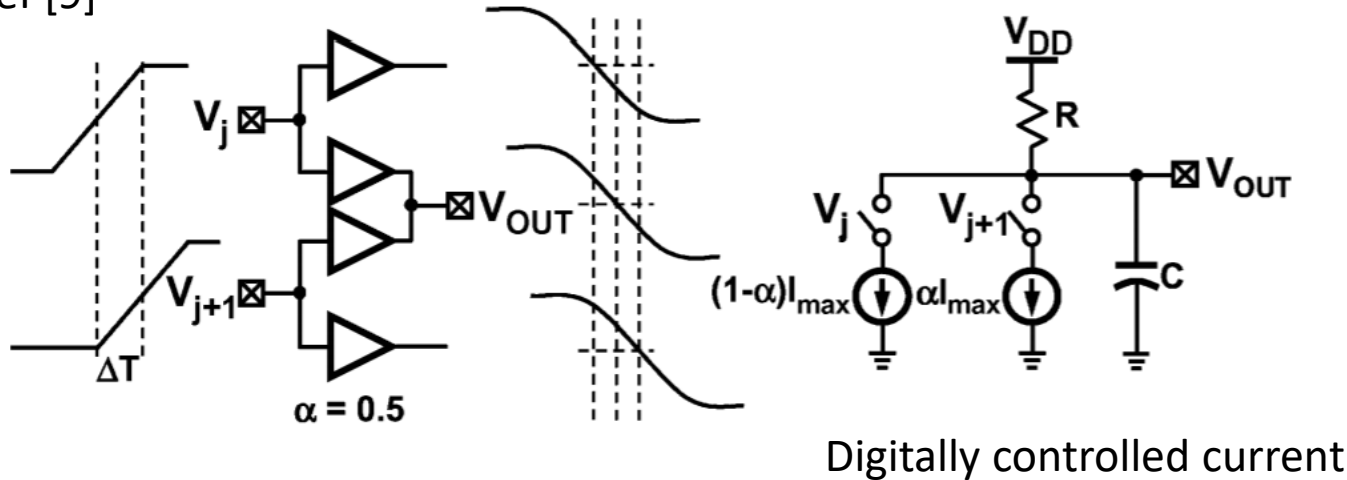


- If all inverters are identical, the output phase $\Phi_{mid} = \frac{\Phi_1 + \Phi_2}{2}$ equals to the average of the input phases.
- Multiple levels of phase interpolation to achieve finer resolution
- Mismatch causes nonlinearity or even non-monotonicity.



DTC Implementation – Phase Interpolation

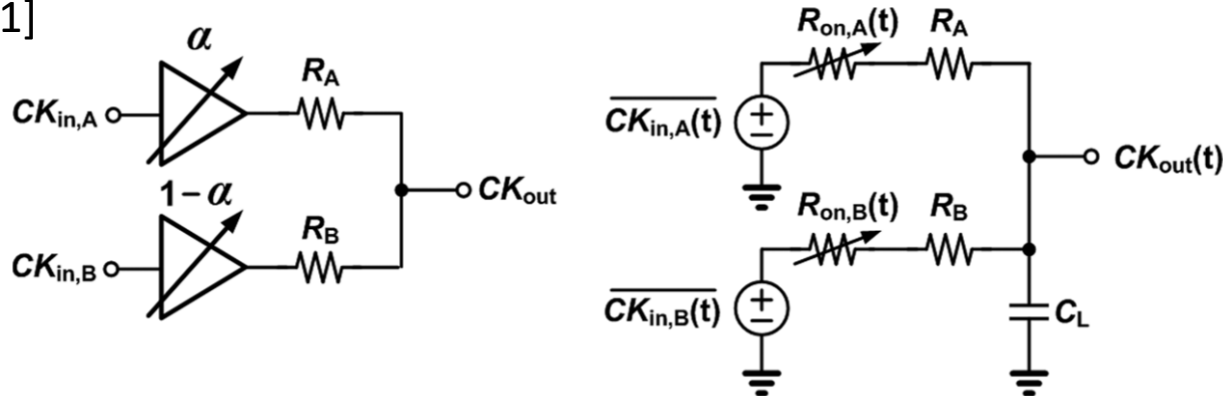
Ref [9]



Digitally controlled current phase interpolation

- V_j and V_{j+1} time difference ΔT
- If $\alpha=0$, V_{out} falling edge time is $t_0 = \frac{C}{I_{max}} (V_{DD} - V_{th})$
- If $\alpha=1$, V_{out} falling edge time is $t_1 = \frac{C}{I_{max}} (V_{DD} - V_{th}) + \Delta T$

Ref [11]



For $0 < \alpha < 1$

$$V_{DD} - \frac{I_{max}(1-\alpha)t_\alpha}{C} - \frac{I_{max}\alpha(t_\alpha - \Delta T)}{C} = V_{th}$$

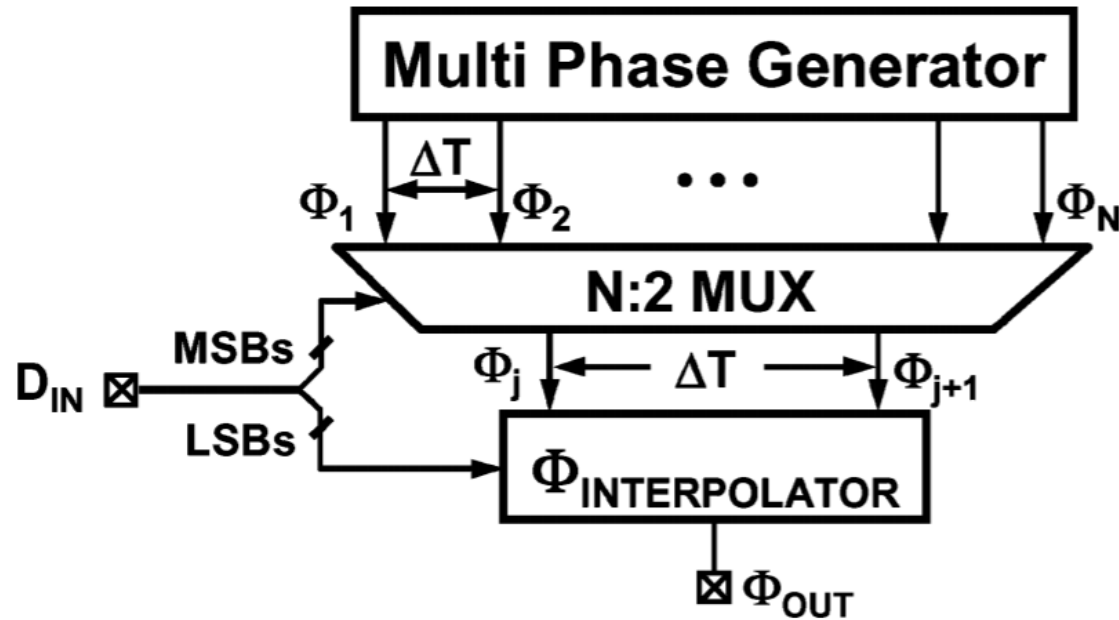
$$\frac{I_{max}}{C} (t_\alpha - \alpha\Delta T) = V_{DD} - V_{th}$$

$$t_\alpha = t_0 + \alpha \cdot \Delta T$$

Ref [9]



DTC Implementation – Delay Line + Phase Interpolation

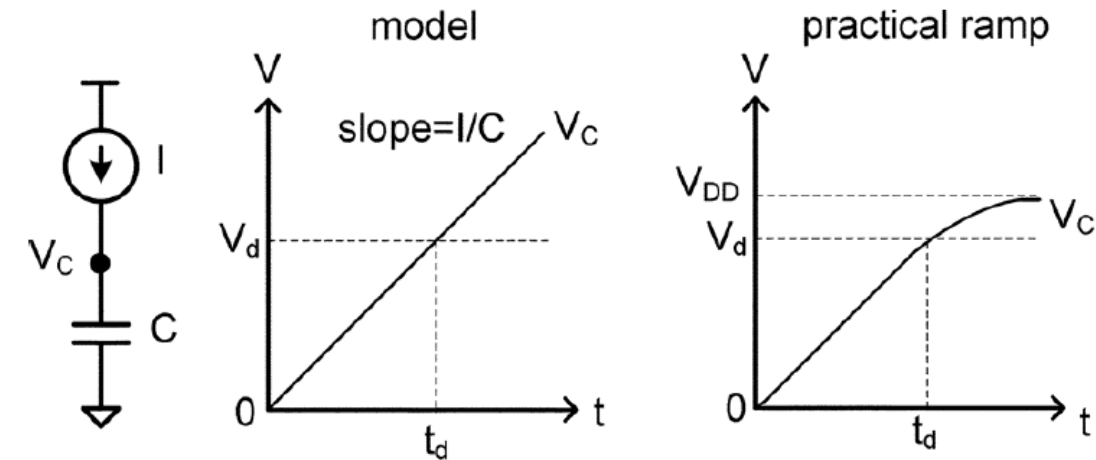


- Phase interpolator (PI) input phase difference usually small
 - If inputs are 180° out of phase, it may not be functional
 - Range is limited to $\sim \pi/4$
- Delay line based DTC generate output phase range of 2π
 - Resolution limited by one unit delay
- Delay line based DTC outputs as phase interpolator inputs
 - Full range 2π by delay line
 - Fine resolution by phase interpolation

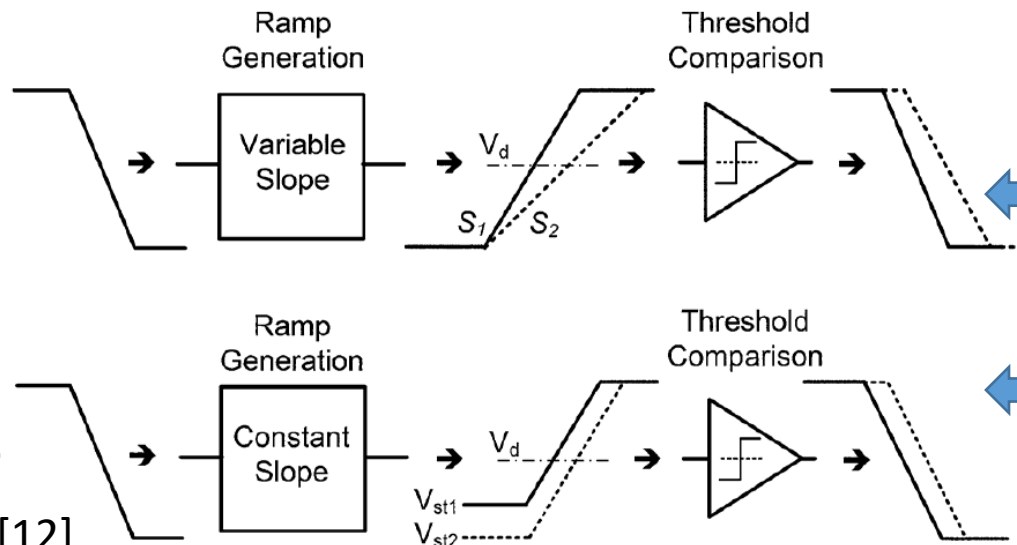
Ref [9]



DTC Implementation – Voltage Ramp Generation



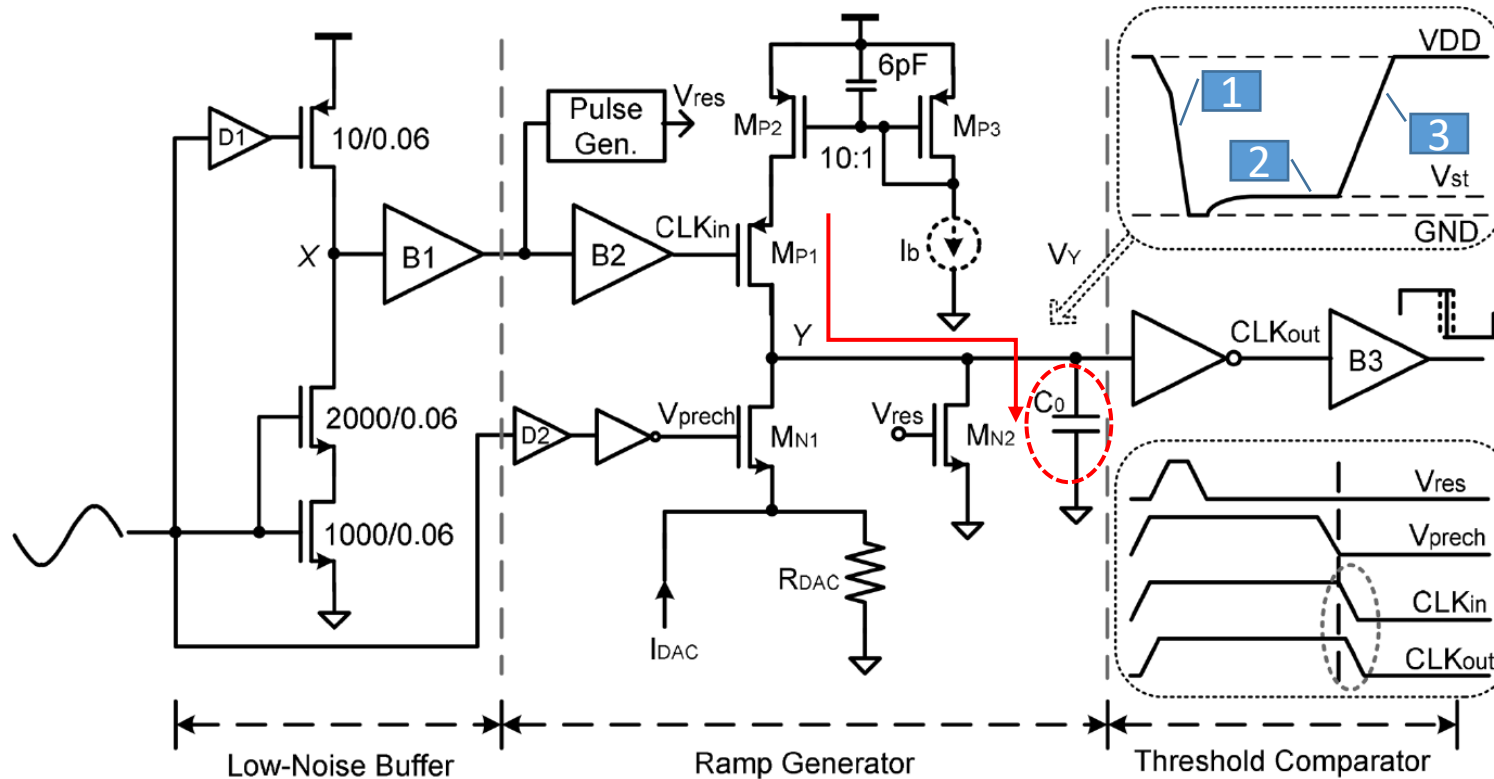
- Generate a voltage ramp by a current source charging a capacitor
- A comparator with threshold voltage V_d generates output signal edge
- Delay t_d determined by the voltage ramp slope and the delta voltage ($V_d - V_{start}$)
- Change t_d by changing voltage slope (variable slope DTC)
- Change t_d by changing T_{start} (constant slope DTC)



Ref [12]



DTC Implementation – Constant Slope DTC

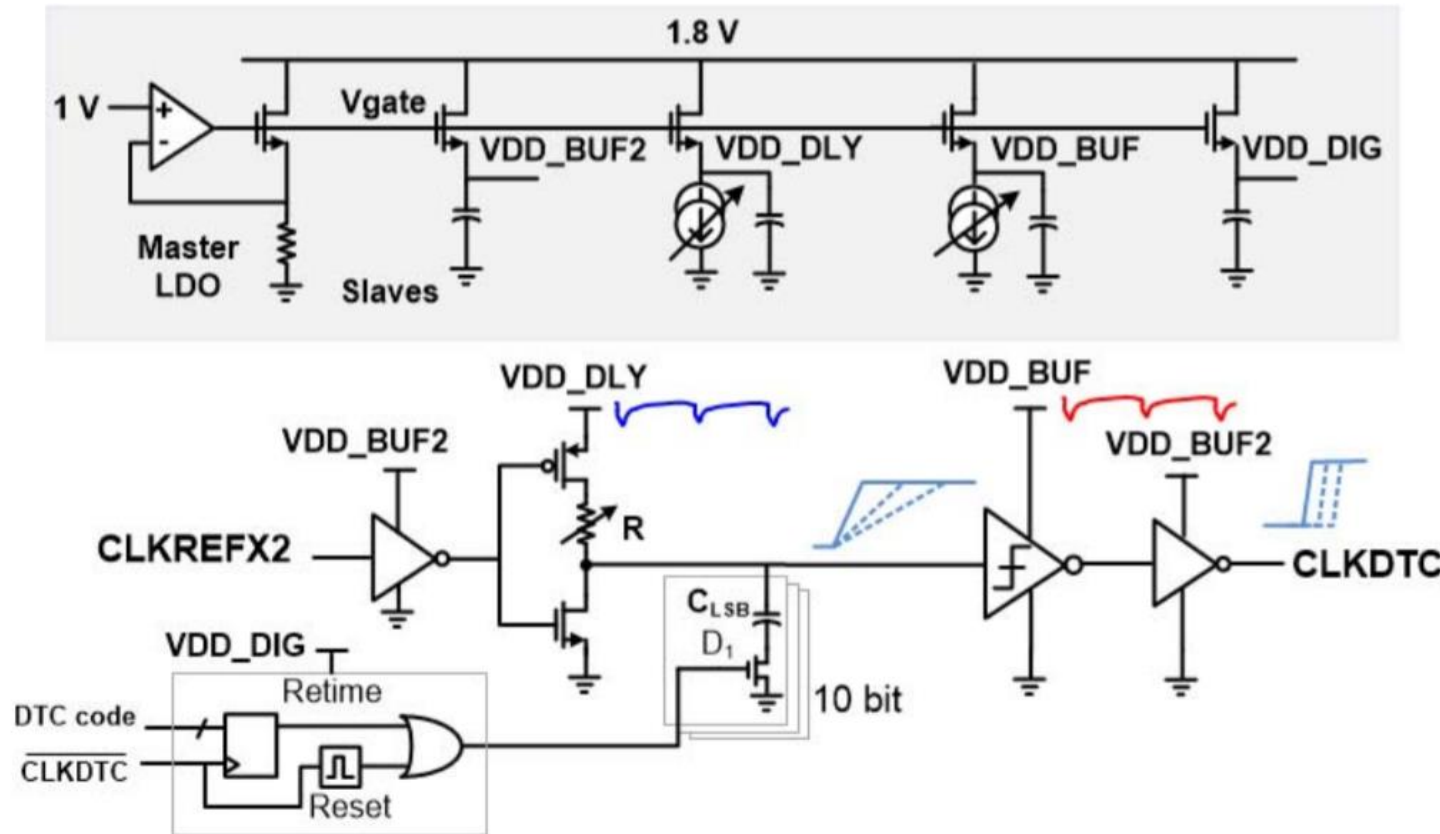


- $t_d = \frac{V_d - V_{st}}{I_{ch}/C_0}$
- In each DTC cycle
 - 1) C_0 is reset to GND by M_{N2}
 - 2) Pre-charge to V_{st} through M_{N1}
 - 3) Charge at a constant slope with a current of $10 \cdot I_b$ through M_{P1}
- Pre-charged voltage V_{st} is controlled by DAC $V_{st} = I_{DAC} \cdot R_{DAC}$

Ref [12]



DTC Implementation – Voltage Ramp Generation with RC



$$V_C = V_{DD} \left(1 - e^{-\frac{t}{RC}}\right) \quad (1)$$

Time delay for V_C to reach inverter threshold voltage V_{th} :

$$t_d = RC \ln \left(\frac{V_{DD}}{V_{DD} - V_{th}} \right) \quad (2)$$

If $V_{th} = 0.5V_{DD}$,

$$t_d = RC \cdot \ln 2 \quad (3)$$

- By digitally changing C , t_d can be changed
- Although RC charge voltage over time is nonlinear (Eq. (1)), t_d vs C is linear.

- Current source flicker and thermal noise degrades DTC noise performance
- Replace the current source with a resistor from VDD – less noise generation
- Voltage ramp slope determined by RC time constant
- RC time constant is digitally controlled by programming the capacitors

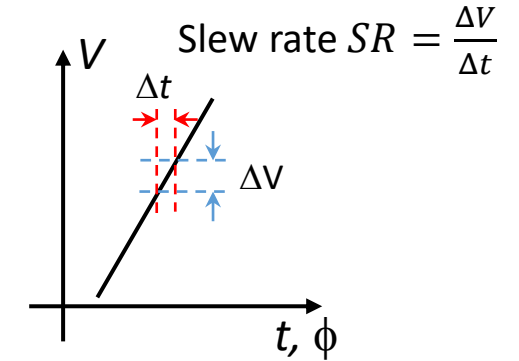
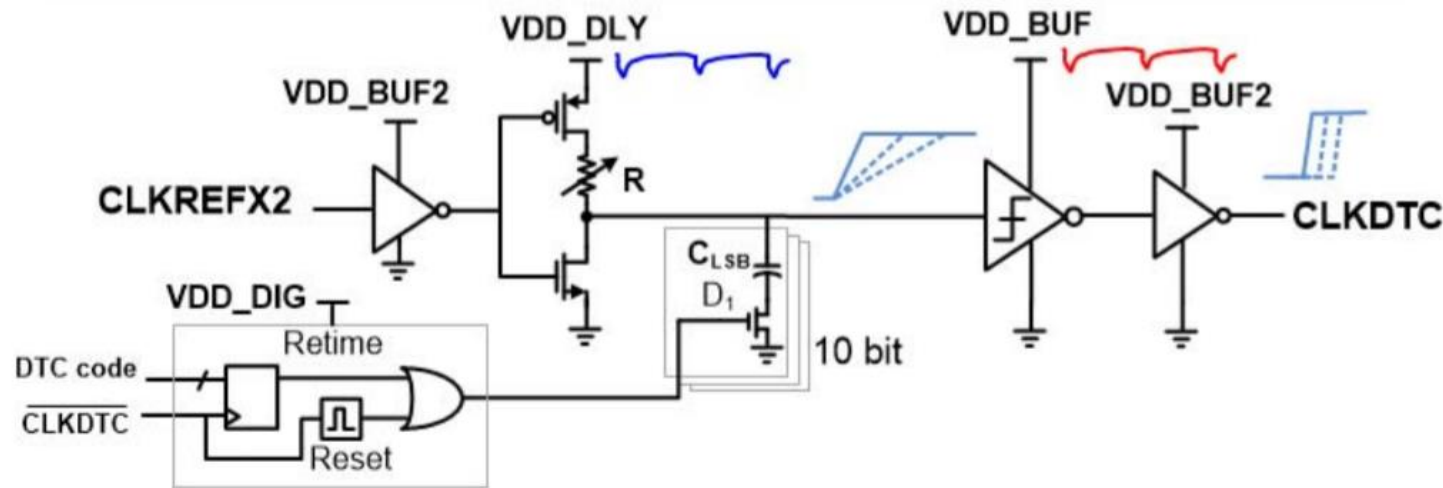


DTC Performance Requirements

- Three main DTC applications
 - Reducing DSM quantization noise in APLL and DPLL
 - Reducing TDC range in DPLL
 - Implementing fractional frequency synthesis in BBPLL and sampling/sub-sampling PLL
- DTC resolution – quantization noise
- DTC range – DSM quantization noise range, a few VCO periods
- DTC noise, circuit intrinsic flicker and thermal noise
 - As part of reference path noise, DTC noise contributes to PLL in-band phase noise (IPN)
- DTC gain
 - $G_{DTC} = \frac{t_d}{D_{in}} = t_{res}$
 - An absolute time delay needs to be generated
 - DTC gain accuracy is important and requires calibration
- DTC linearity, measured by INL or DNL, is important to DSM noise folding



DTC Performance Analysis Example



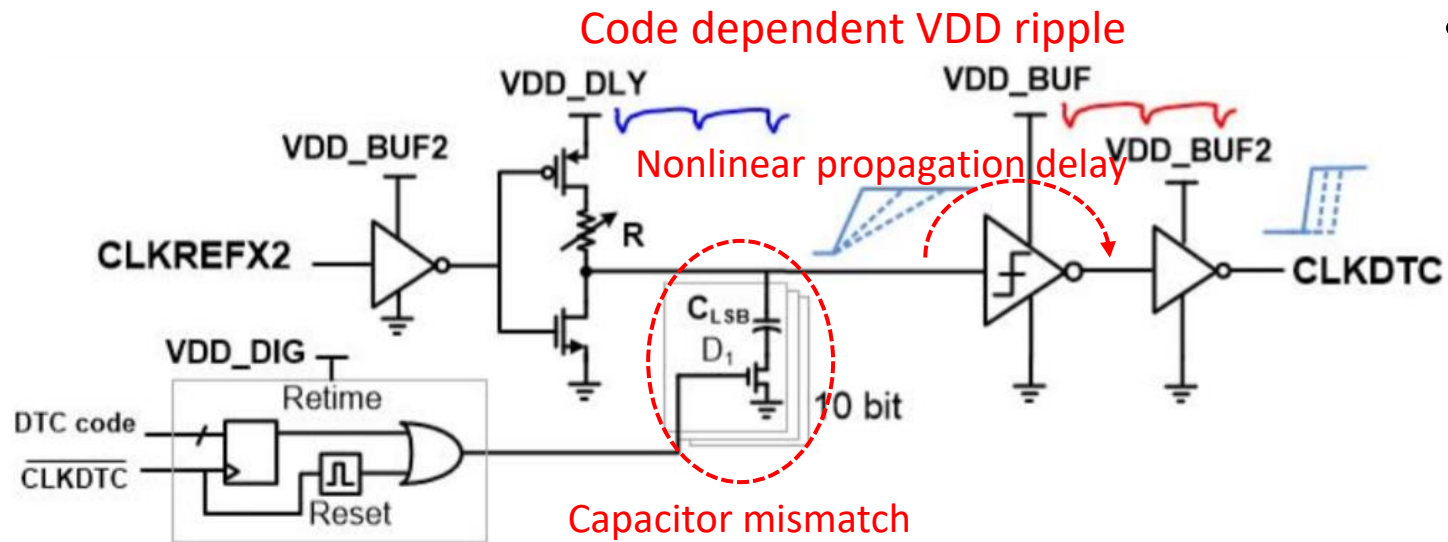
$$\overline{\Delta V^2} = 4kTR$$

$$\overline{\Delta \phi^2} = \left(\frac{2\pi}{T_{ref}} \right)^2 \overline{\Delta t^2} = (2\pi f_{ref})^2 \frac{4kTR}{SR^2}$$

- Resolution: $T_{res} = RC_{LSB} \cdot \ln 2$
- Range: $T_{total} = 2^{10} T_{res}$
- Thermal noise: $\mathcal{L} \cong 10 \cdot \log_{10} \left[\frac{1}{2} \left(\frac{2\pi \cdot f_{ref}}{SR} \right)^2 \cdot 4kTR \frac{ENBW}{f_{ref}/2} \right]$
 - ENBW: effective BW of RC delay stage, $ENBW = \frac{1}{4RC}$
 - Slew rate is DTC code dependent. The thermal noise is the worst at maximum code where the delay is the largest.
- Quantization noise: $\mathcal{L} \cong 10 \cdot \log_{10} \left[\left(\frac{2\pi}{\sqrt{12}} \cdot \frac{T_{res}}{T_{VCO}} \right)^2 \frac{1}{f_{ref}} \right]$



Sources of DTC Nonlinearity

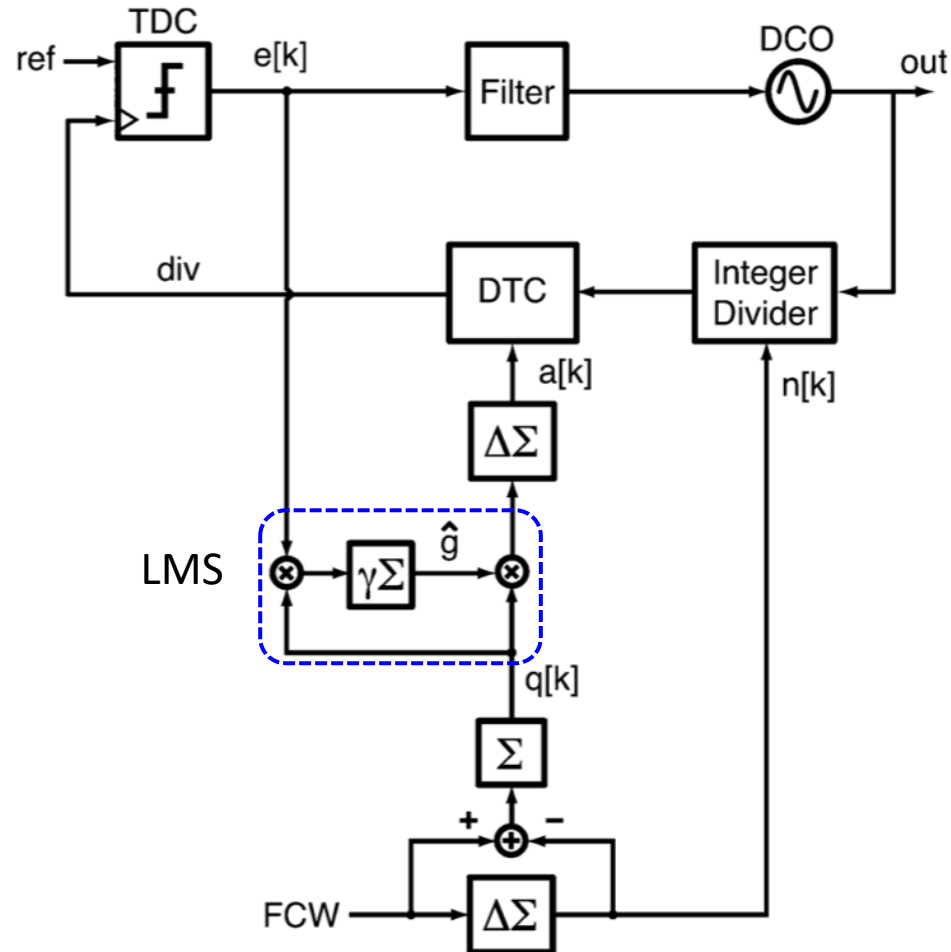


- Dynamic nonlinearity – code dependent VDD ripple
 - RC charge and discharge, inverter buffer turning on and off all cause VDD supply settling.
 - With finite VDD settling time, VDD does not reach its steady state by the time the next event (code) comes.
 - Settling of the current event and the previous event are both code dependent and both impact the DTC delay.

- Static nonlinearity – capacitor array mismatch
 - $t_d = RC \cdot \ln 2$, $T_{res} = RC_{LSB} \cdot \ln 2$
 - Mismatch in capacitor array causes nonlinearity
 - Dynamic element match (DEM) improves linearity due to mismatch
- Static nonlinearity – slope dependent inverter propagation delay
 - $t_d = RC \cdot \ln 2$ is linear over C or control code. However, the following inverter propagation delay is input slope dependent, which is code dependent.



DTC Gain Calibration

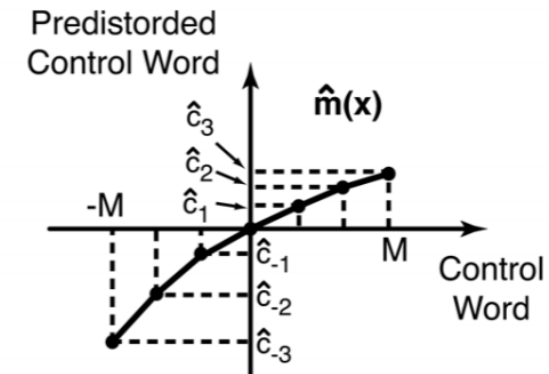
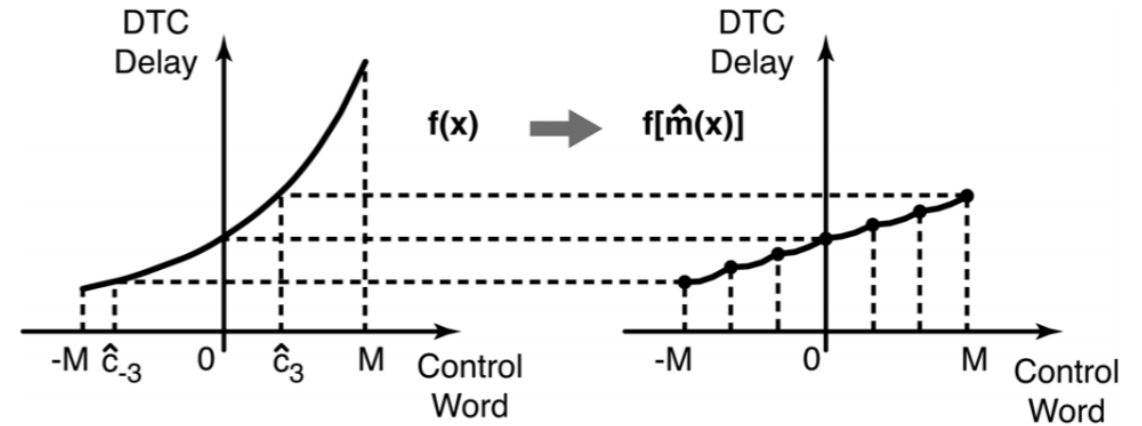
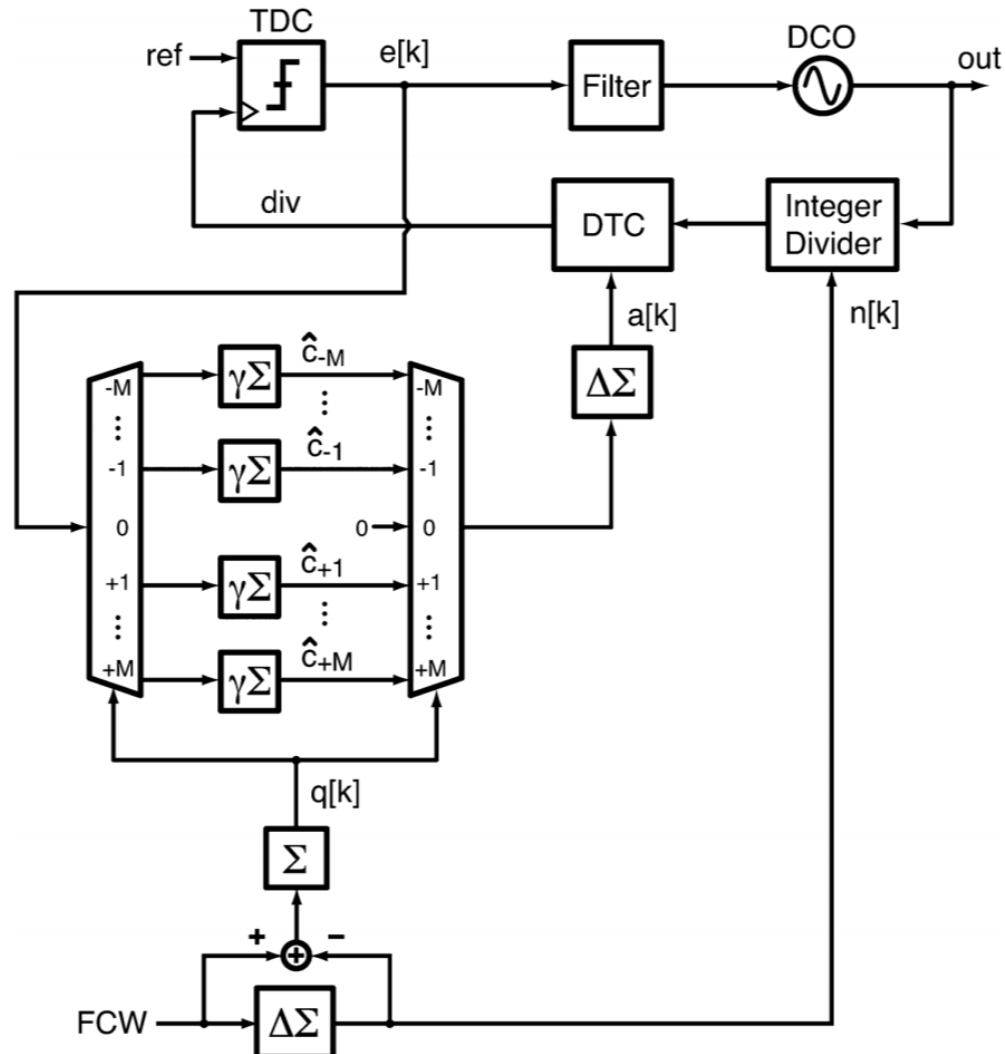


- DTC gain calibration
 - DTC generates a time delay
 - The absolute delay time is important
 - DTC gain is calibrated using VCO period
 - Least mean square (LMS) algorithm
- DTC linearity calibration
 - Look-up table approach, segmented gain calibration
 - Time consuming ($\sim ms$)

Ref [14]



DTC Linearity Calibration



Summary

- Applications of DTC
 - In fractional-N PLL, DTC reduces DSM quantization noise
 - In fractional-N digital PLL, DTC reduces TDC range requirement
 - In BBPLL, subsampling PLL and sampling PLL, DTC enables fractional-N synthesis
- DTC topologies
 - Delay line-based DTC
 - Phase interpolation DTC
 - Voltage ramp generator-based DTC
 - Constant slope DTC
 - RC delay-based DTC
- DTC performance
 - Range, resolution, gain
 - Noise
 - Linearity



References

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